# Yusef Syed

**Software Engineer: AI agent security & evals | 14 merged PRs: Microsoft, Apache Arrow, W&B**

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Toronto, ON | U.S. & Canadian citizen, no sponsorship needed | Will relocate to San Francisco or New York

**Available full-time from January 2027:** security engineer, evals engineer, AI red team, AI safety research engineer

## Open Source

**14 merged PRs** (AI-assisted): Microsoft, Apache Arrow, W&B, Inspect Scout, Kornia | Full list **Sep – Oct 2026**

- Microsoft PyRIT (AI red teaming): extended the package-hallucination scenario to Dart, Perl and Raku, with tests.
- W&B RAI Toolkit: landed five evaluation fixes, including keeping a failed prediction's error text from scoring as a pass.
- Apache Arrow: extended four PyArrow slice APIs to accept Arrow integer scalars, with regression tests.
- Microsoft Agent Lightning: landed four reliability fixes, e.g. case-insensitive auth-header replacement.

## Projects

**ToolEffectBench** | *Python, Inspect, Docker* | github.com/YusefSyed/tool-effect-bench | Paper v0.7 (41 pp) **2026**

- Built, with AI coding agents, an Inspect benchmark scoring prompt injection from a sandbox's database state, not transcripts (188 samples, 17 models, 5,628 scored runs); on one synthetic task, agents' final reports denied **13 of 232** recorded forbidden publications.

**effect-tracer** | *Rust, Linux ptrace/seccomp, Ed25519* | github.com/YusefSyed/effect-tracer **2026**

- Built, with AI coding agents, a tracer that signs a process tree's file writes, connects, DNS and execs into a hash chain: **19 of 20** seeded bypasses caught at 1.1–1.9x overhead in seccomp mode (20 of 20 at 5–298x in full mode); 0 undeclared events in 30 clean runs. Signed v0.1.2 release with SBOM and VEX.

## Experience

**Scopehaven | github.com/YusefSyed/scopehaven-oss** **Oct 2026 – Present**

*Software Engineer (creator, solo project); security testing for AI agents*

- Built **Scopehaven**, a Node.js CLI and GitHub Action that tests whether an AI agent touched the wrong account, tenant or refund, decided from the app's own records rather than the agent's report.
- Released the **Apache-2.0 open edition** (v0.7.0: 4,800+ tests, zero dependencies, GitHub Action, MCP test target); on my synthetic test app the vulnerable build fails 26 of 44 checks.
- Designed signed, tamper-evident evidence (DSSE/in-toto exports, a hash-chained ledger) and a stdlib-Python checker that re-derives verdicts from kept records: 88 of 88 on both demo builds (synthetic app, no model), 8 of 8 tampered copies rejected.
- Built it with AI coding agents under a merge gate I ran as the only human on the repo (exact-head CI, a recorded review by a second AI session, secret scanning): 57 review verdicts asked for changes.

**Independent Security Research** **Oct 2026 – Present**

*Security Researcher* *Remote*

- Filed 15 private security advisories and 2 HackerOne reports (authorization, credential and data-exposure flaws) with 10 open-source projects, each reproduced 3 of 3 with controls first by an automated harness; all awaiting maintainer triage.
- Ran an internal pilot on an open-source AI workspace (synthetic data): after a user was suspended, an in-flight agent run still wrote a file in 2 of 14 cases; a partial local patch passed 14/14 on the tested path.

## Education

**University of Toronto** **Expected May 2030**

*B.Sc., Math & Physical Sciences (first year); intended Math + CS; would leave for the right role*

**Coursework:** Linear Algebra, Proofs & Calculus, Statistics; Data Structures (LCCC, dual enrollment)

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Swift

**AI & security:** application security (AppSec), AI security, AI red teaming, prompt-injection evals, vulnerability reproduction, access-control testing (IDOR, RLS), Inspect, PyRIT, OWASP LLM/Agentic Top 10, MITRE ATLAS

**Systems & tools:** Node.js, React Native, SwiftUI, PostgreSQL, Docker, GitHub Actions, PyTorch, Claude Code

## Other Projects & Activities

**Providence** | *Swift* | **1st Place, Aramco Americas Challenge** (Best Beginner Hack), Hack the North **Sep 2026**

**Aesthetics AI** & **Tiraz** | *TypeScript, React Native* | two iOS apps on the **App Store** | Study **2026**

**USA Powerlifting (60 kg, 2025):** #2 in PA, #19 nationally | **Feeding the Valley**, VP: $20,000 for food banks